

Introduction

- Creatinine is a biomarker widely used to assess kidney function and is strongly associated with morbidity and mortality, motivating our investigation of creatinine levels.¹
- The renal system regulates blood filtration and fluid homeostasis; thus, we selected 25 candidate variables focused on factors closely related to these functions. Including summary statistics of last 72 hours for time-varying covariates increased the number of columns to 60.²

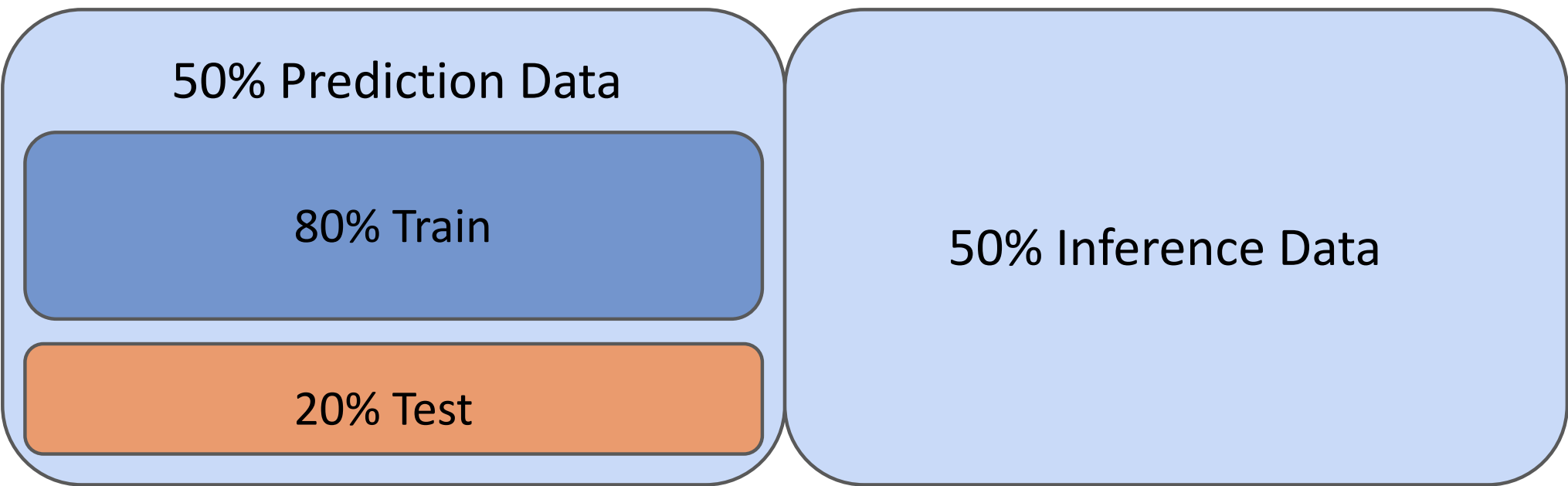
Research Question

Which demographic characteristics, comorbidities, medications and laboratory biomarker are most important in predicting serum creatinine among adult ICU patients prior to dialysis? For interpretability, how are these variables associated with serum creatinine?

Data

ICU creatinine measurements from adult patients prior to first dialysis, extracted from the MIMIC-IV database (Beth Israel Deaconess Medical Center, Boston, 2008-2019). The dataset was subsequently divided into two equal subsets using a 50/50 split, stratified by race and gender; it contained **54,574 patients with 317,980 creatinine observations**, with a median of 3, 25th percentile of 2, and 75th percentile of 6 per patient. Missingness was even across race and gender, with highest missingness in pH (31%) and lactate (40%).

The similar trajectory patterns observed across Chronic Kidney Disease (CKD) stages within each row supports grouping them into 4 categories according to disease severity.



Methods

To identify variables most strongly associated with serum creatinine trajectories, we began by building random forest, XGBoost, and feed-forward neural network models. We applied **SHapley Additive exPlanations (SHAP) values and permutation-based importance methods** to quantify and extract which variables were most important in each model's predictions. Using multiple models allowed us to assess whether the important variables identified were consistent across different approaches, which helped us confirm the consistency of our findings.

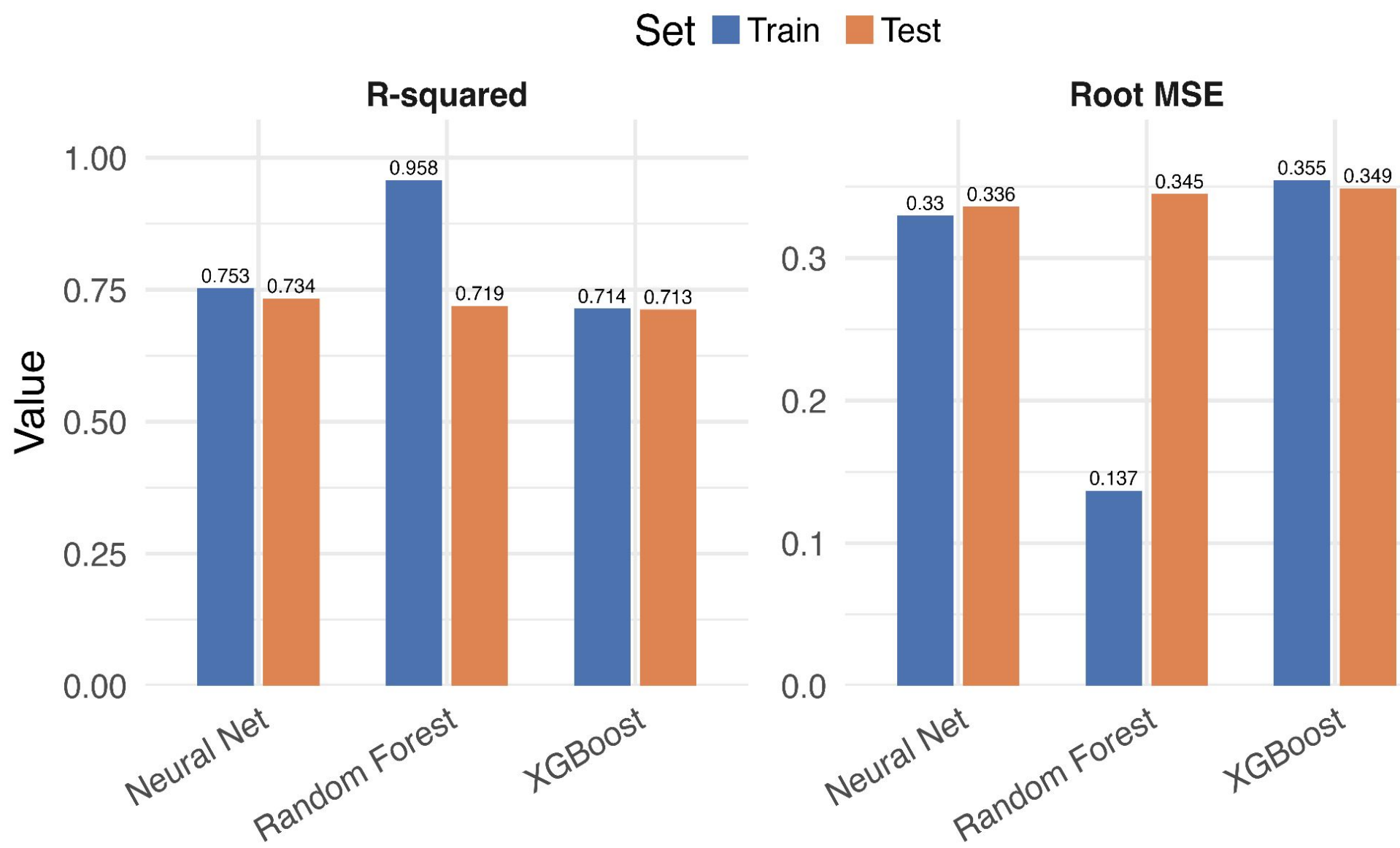


Figure 2: Model Performance: Train vs. Test

Using the first 50% split, we trained and tested all three methods on an 80/20 test and train split, and all of them performed similarly based on evaluation metrics (R², Root Mean Squared Error).

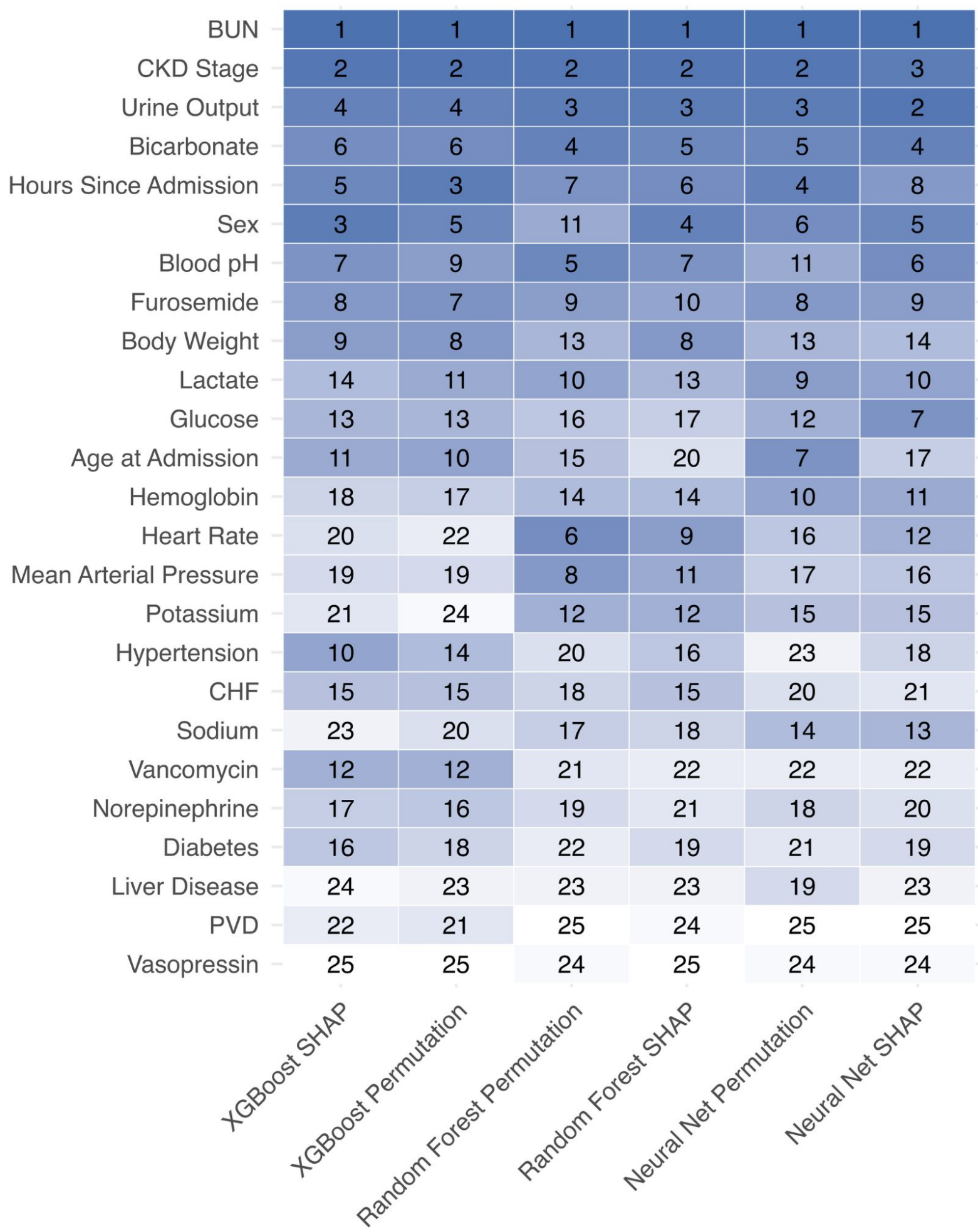


Figure 3: Variable Importance Rank Heatmap

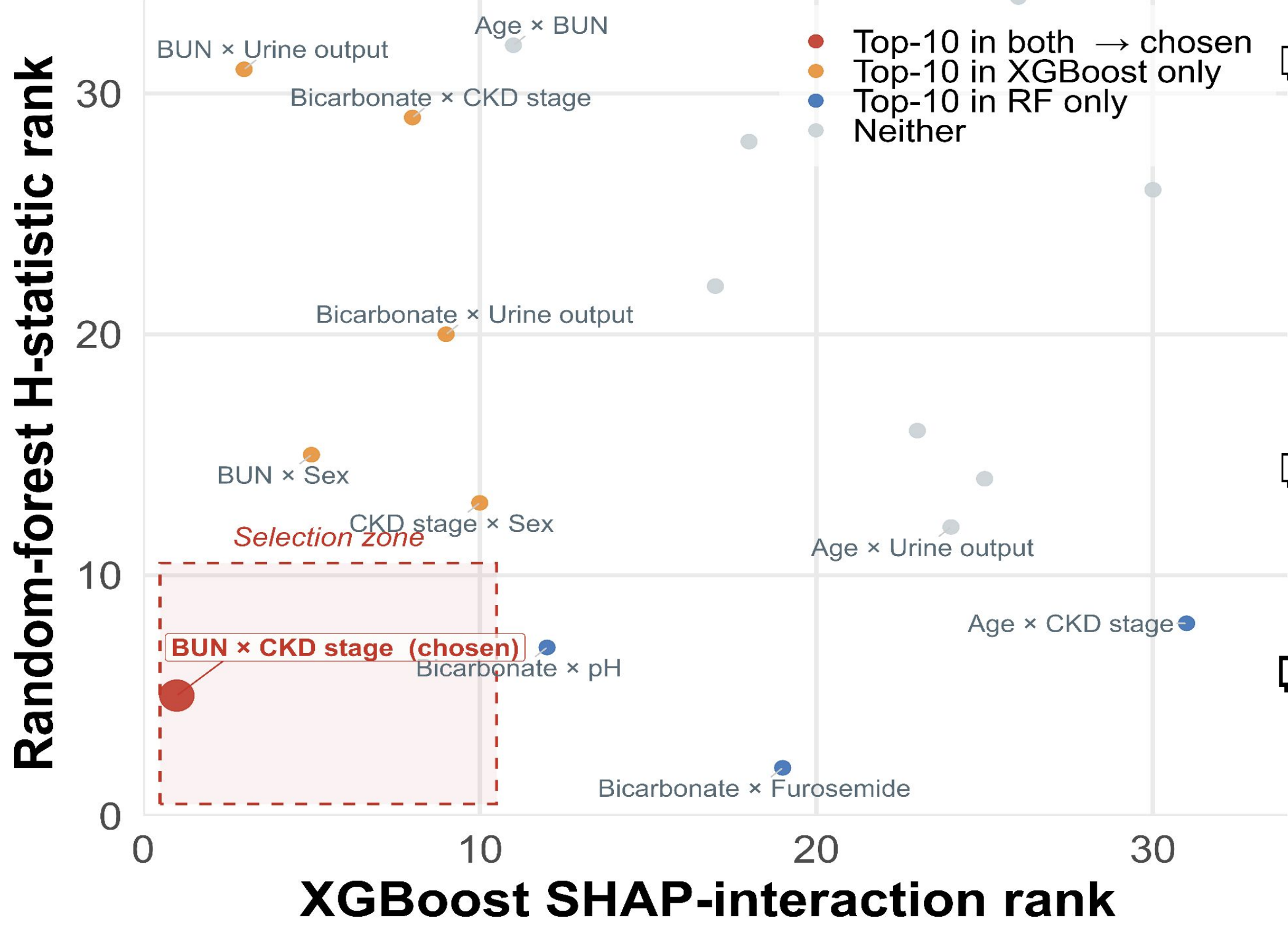


Figure 4: Choosing the Interaction Term by Cross-Learner Agreement

Results

Variable	Adjusted Coefficient (x100)	Adjusted p-value	Unadjusted Coefficient (x100)	Unadjusted p-value	Unadjusted FMI (Adjusted)
Age at admission	0.18	<0.001	0.89	<0.001	0.04
Bicarbonate (lowest value)	-0.97	<0.001	-1.56	<0.001	0.38
BUN (most recent value)	0.87	<0.001	1.03	<0.001	0.3
BUN x CKD stage: Mild (interaction)	-0.42	<0.001	N/A	N/A	0.45
BUN x CKD stage: Serious (interaction)	-0.76	<0.001	N/A	N/A	0.44
BUN x CKD stage: Unspecified (interaction)	-0.32	<0.001	N/A	N/A	0.45
CKD stage: Mild (vs. None)	57.32	<0.001	65.21	<0.001	0.46
CKD stage: Serious (vs. None)	160.04	<0.001	153.15	<0.001	0.1
CKD stage: Unspecified (vs. None)	55	<0.001	67.27	<0.001	0.4
Furosemide (total dose)	0.03	<0.001	0.06	<0.001	0.02
Sex: Female (vs. Male)	-18.28	<0.001	-24.88	<0.001	0.01
Glucose (peak value)	<0	0.19	0.02	<0.001	0.3
Heart rate (peak value)	0.01	0.42	-0.05	<0.001	0.27
Hemoglobin (lowest value)	-0.37	0.03	-1.25	<0.001	0.56
Time since ICU admission	-0.05	<0.001	-0.05	<0.001	0.01
Hypertension (vs. No)	5.79	<0.001	25.97	<0.001	0.01
Lactate (peak value)	0.72	<0.001	1.43	<0.001	0.36
Mean arterial pressure (lowest value)	-0.11	<0.001	-0.19	<0.001	0.42
Blood pH (lowest value)	-24.04	<0.001	-55.73	<0.001	0.52
Urine output (rate, mL/kg/hr)	-0.04	0.15	-0.14	0.05	0.36
Body weight	0.23	<0.001	0.38	<0.001	0.01

Table 1: GEE Summary (Adjusted vs Unadjusted)

- Ranked all 25 candidates across those 6 methods.
- BUN, CKD stage, and urine output rank consistently in the top 4 across every model.**
- Selection rule: a variable is kept if it ranks top-10 in any single method - retained 16 variables for the GEE.
- Age, sex, and weight forced in regardless of rank, as standard demographic information.
- For time-varying variables with multiple candidate summaries (last, min, max, etc.), used the summary that 4 or more models agreed on; otherwise defaulted to the clinical standard.

- Ranked all variable pairs by two different methods: XGBoost SHAP-interaction values and Random Forest H-statistic.
- Selection rule: a pair is kept only if it ranks top-10 in both methods.
- Only BUN x CKD stage qualified.**

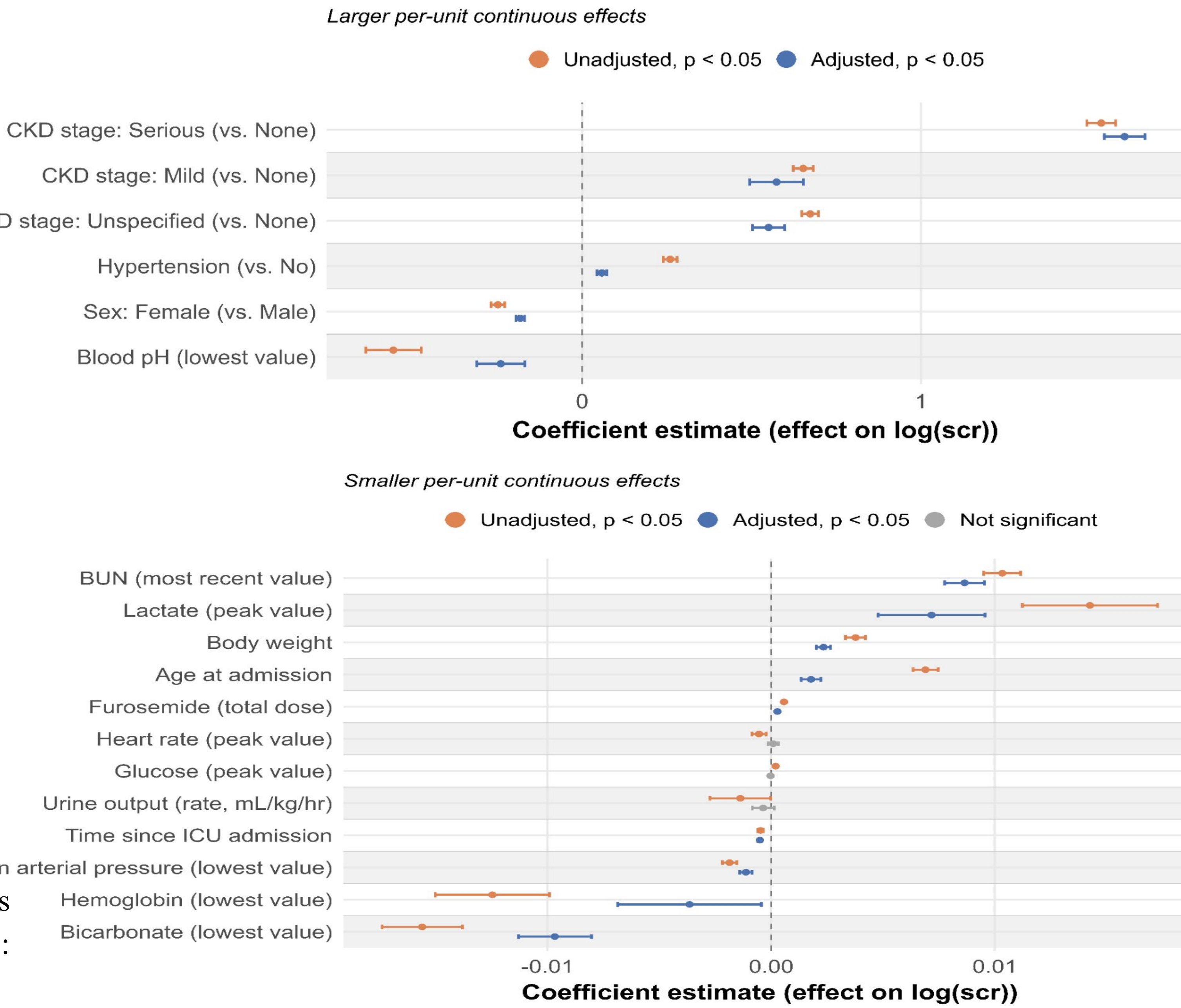


Figure 5: Adjusted vs. Unadjusted GEE Coefficients

Creatinine trajectory in critically ill adults is primarily associated with **chronic kidney status** rather than acute clinical variables. Adjustment attenuates most coefficients, and heart rate and glucose lose significance entirely, though the BUN x CKD interaction suggests acute markers matter more when baseline kidney function is preserved.

Discussion

- Our design assumes missingness at random (MAR) for all missing data.
- Due to limited computing power, we were able to run multiple imputations only 10 times³.
- This project is solely an association analysis, with no causal claims made.
- The analyses were restricted to linear associations; therefore, the exact functional form of the relationships remains unknown.
- We tried using the more ideal machine learning models with random effects, but they resulted in poor predictive power.

Future directions include:

- Creating a causal analysis of important variables that lead to Acute Kidney Injury.
- Using exploratory visualization and/or flexible regression methods to examine functional relationships.
- Examining whether the influence of key variables are associated with downstream outcomes such as ICU stay lengths, long-term mortality and kidney function outcomes.

Acknowledgements

Thank you to our mentors Dr. Henan Xu, Dr. Cheng-Han Yang, and Waveley Qiu. We would also like to thank Dr. Bhramar Mukherjee, Jackson Higginbottom, Yiren Hou, Amara Neal, and BDSY for the opportunity.

References:

